Experiment - 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

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Syed Faiyaz Ahamed
241401113 B.Tech CSBS

AIM

To implement and explore the security, access-control, data-sharing and governance features of a Zoho Creator application.

INTRODUCTION

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an essential part of application design.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations a user may perform on the components of an application. Role-based access places users in a role hierarchy, which influences the records they can see. Data sharing rules extend the visibility of records from one role or user to another at a defined access level. Administrative settings allow the application to be configured and managed. Together they ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use.

Application Used

The practical was carried out in Zoho Creator using the Hotel Booking application, which is used to manage room reservations and guest details. It served as the working environment for exploring the security and governance features described in this record. A hotel is a suitable example because different people, such as management, front-desk staff and guests, need different levels of access to the same booking data.

Security and Governance Architecture

Figure 1 is a conceptual diagram of the security and governance structure of the Hotel Booking application. It is not a screenshot of the Zoho Creator interface.

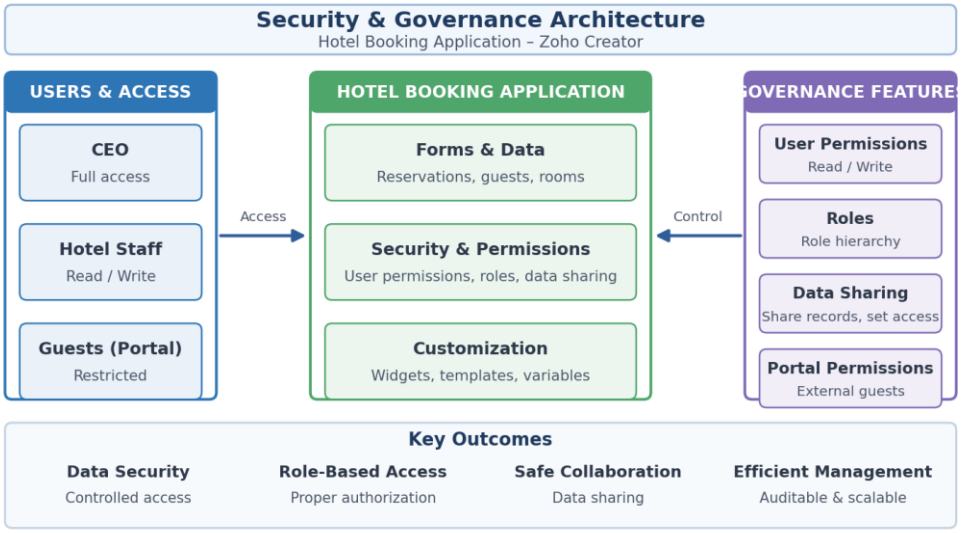


Figure 1: Security & Governance Architecture (Conceptual Diagram)

The diagram shows users (such as the CEO, hotel staff with Read / Write access and guests using a portal) accessing the application, which contains forms and data, security and permissions, and customisation items. The governance features on the right (user permissions, roles, data sharing and portal user permissions) manage and control that access. The key outcomes are data security through controlled access, role-based access, safe collaboration through data sharing and efficient management.

PROCEDURE

1. Open the Hotel Booking application in Zoho Creator in Edit mode.
2. Open Settings from the application editor and select User Permissions.
3. Review the Permissions tab and the predefined permission sets.
4. Open the Roles tab and review the role hierarchy.
5. Use Add Role to define a role for hotel management and review its settings.
6. Open the Data Sharing tab and create a new share rule for the Hotel Booking form.
7. Open Portal User Permissions and review the options for external users.
8. Record the observations for each feature.

User Permissions

User Permissions is opened from Settings in the application editor. It contains the tabs Permissions, Roles and Data Sharing, and is used to manage the permissions and roles of users and the visibility of records for each user. The Permissions tab lists the permission sets available in the application and provides an Add Permission option. Two predefined permissions are provided:

Name	Description (as displayed)
Read	This profile will have read permission for all components
Write	This profile will have write permission for all components

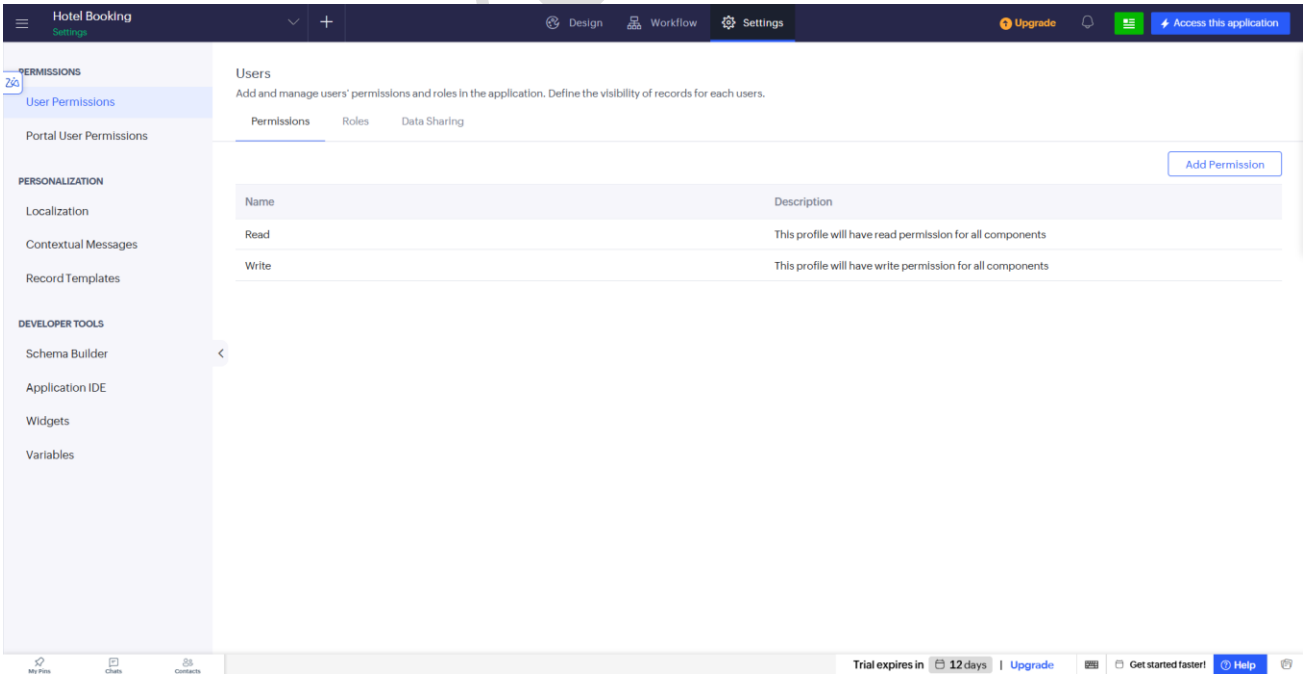


Figure 2: User Permissions in the Hotel Booking Application

Role-Based Access Control

The Roles tab shows the role hierarchy of the application. By default the CEO role is displayed, and the options Expand All, Collapse All, Manage Role Hierarchy and Add Role are available. A role placed higher in the hierarchy can see the records of the roles that report to it. A suitable hierarchy for the Hotel Booking application is shown below.

Role	Reports To	Typical Responsibility
CEO	– (top of hierarchy)	Overall access to all bookings and reports
Hotel Manager	CEO	Manages reservations, rooms and staff
Front-Desk Staff	Hotel Manager	Creates and updates reservations, check-in and check-out
Housekeeping Supervisor	Hotel Manager	Views room status and updates cleaning status

Table 1: Role Hierarchy Planned for the Hotel Booking Application

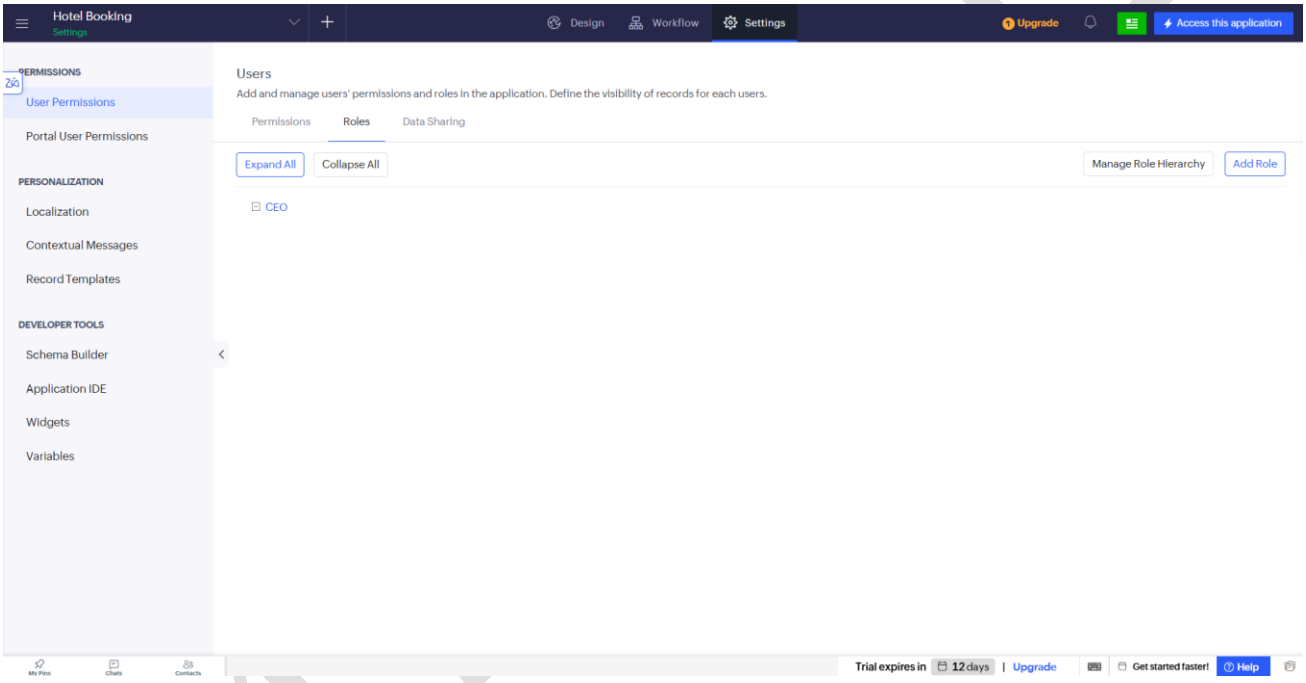


Figure 3: Roles and Role Hierarchy in the Hotel Booking Application

The Add Role option was used to define a role with the following details:

Field	Value
Role Name	Hotel Manager
Reports To	CEO
Description	Manages hotel room reservations
Share Data with Peers	Not selected

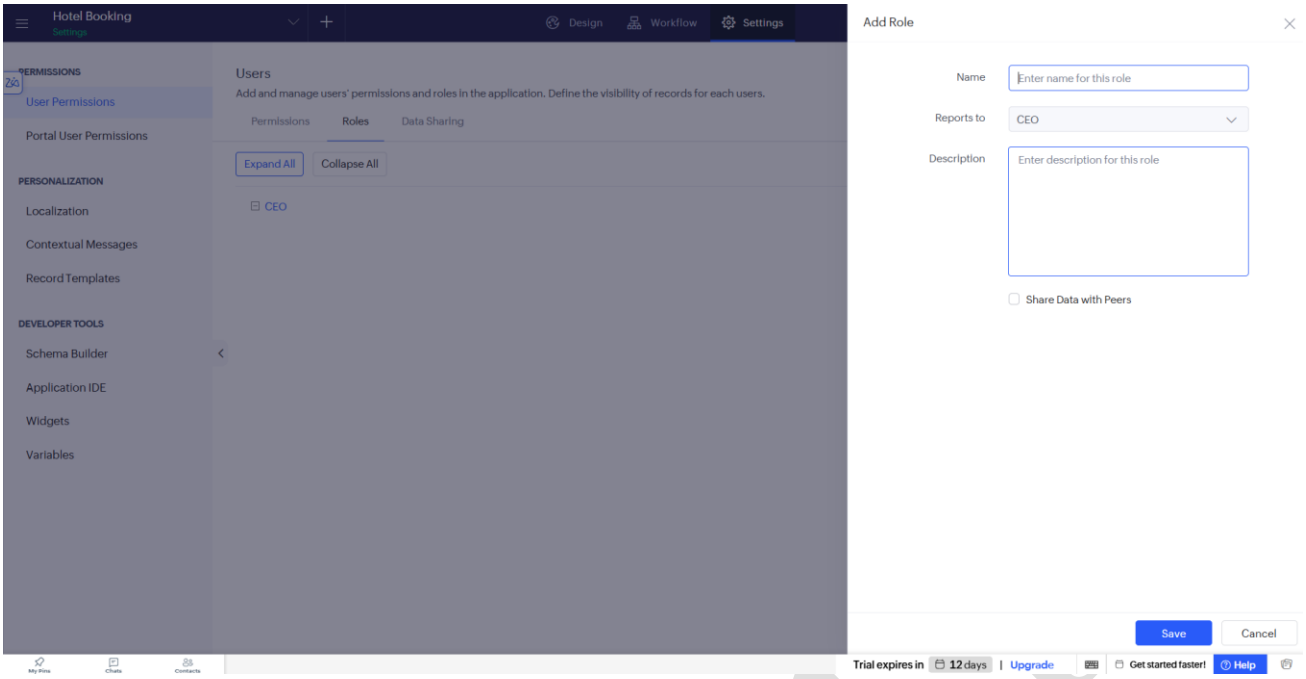


Figure 4: Add Role Panel – Hotel Manager Role Configuration

With "Share Data with Peers" left unselected, users in the same role cannot see each other's records, which follows the principle of giving only the access that is needed.

Data Sharing

Data sharing rules control the visibility of the records of a form. A rule specifies the form, the role or user from which records are shared, the role or user to which they are shared and the level of access. Records can therefore be made visible beyond the default visibility of the role hierarchy. A data sharing rule was configured with the following details:

Parameter	Value
Rule Name	Booking Data Sharing
Form	Hotel Booking
Shared From	CEO (role)
Shared To	Syed Faiyaz Ahamed (user)
Access Level	Read/Write

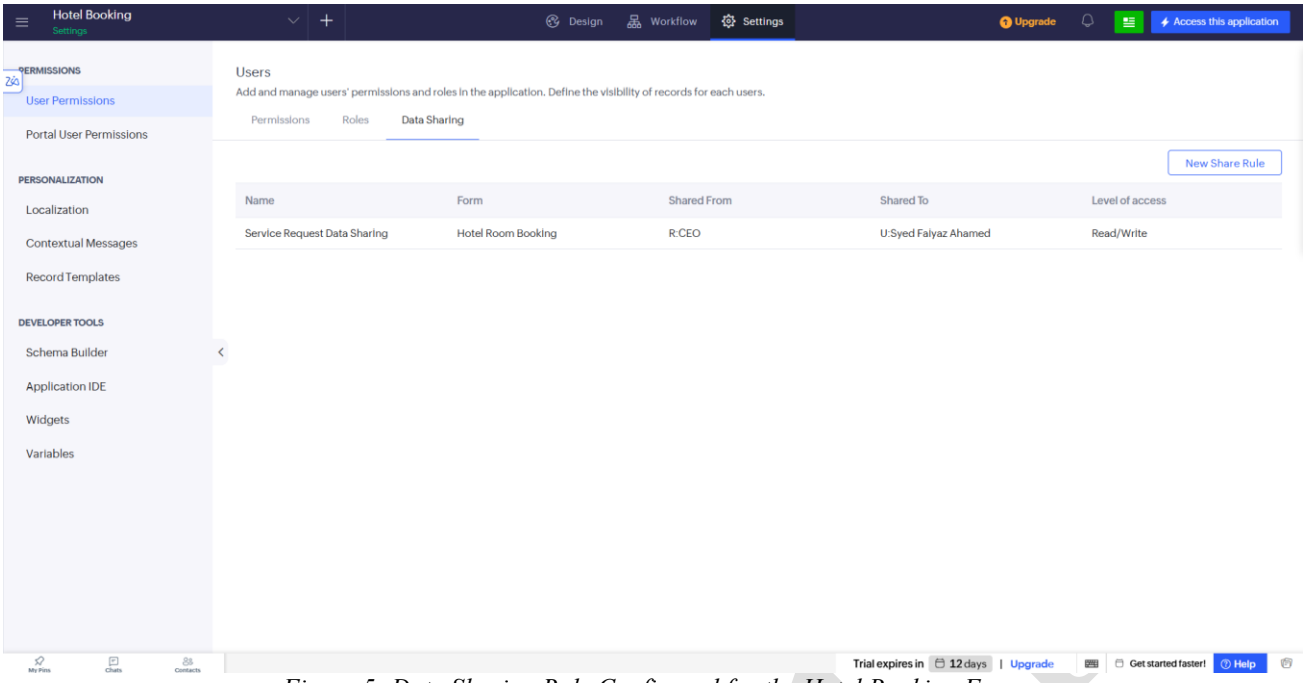


Figure 5: Data Sharing Rule Configured for the Hotel Booking Form

The saved rule appears in the Data Sharing list with the columns Name, Form, Shared From, Shared To and Level of access. With Read/Write access, the recipient can both view and modify the shared records of the Hotel Booking form. The access level decides how much a recipient can do with the shared records, so visibility is extended in a controlled manner.

Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and controls the access of external portal users. In a hotel scenario these are guests who log in to view or manage only their own bookings. This section was opened to examine how access for external users is controlled; the options available depend on the plan of the account.

Security Features and Governance Controls

Figure 6 is a conceptual diagram that groups the security and governance controls of Zoho Creator. It is not a screenshot, and it does not mean that every listed feature was configured in this experiment.

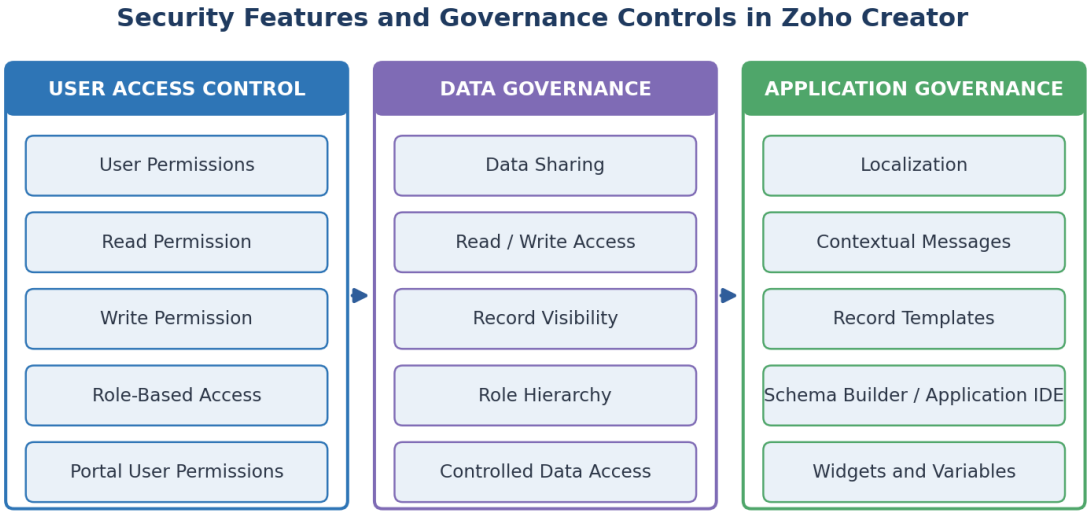


Figure 6: Security Features and Governance Controls in Zoho Creator (Conceptual Diagram)

Control	Explanation
User access control	User permissions (Read and Write), role-based access and portal user permissions decide who can use the application and what they can do
Data governance	Data sharing, read / write access, record visibility and role hierarchy govern how application data is accessed
Application governance	Settings such as localization, contextual messages, record templates, schema builder, application IDE, widgets and variables administer and customise the application
Role hierarchy	Arranges roles in a reporting structure that influences which records a role can see
Data sharing	Extends the visibility of records from one role or user to another at a chosen access level
Administrative controls	Allow an administrator to manage permissions, roles and application settings

Table 2: Security and Governance Controls

Role-Based Access and Data Sharing Flow

Figure 7 shows the security flow in principle: User → User Permissions → Role → Data Sharing Rule → Access Level → Hotel Booking Data.

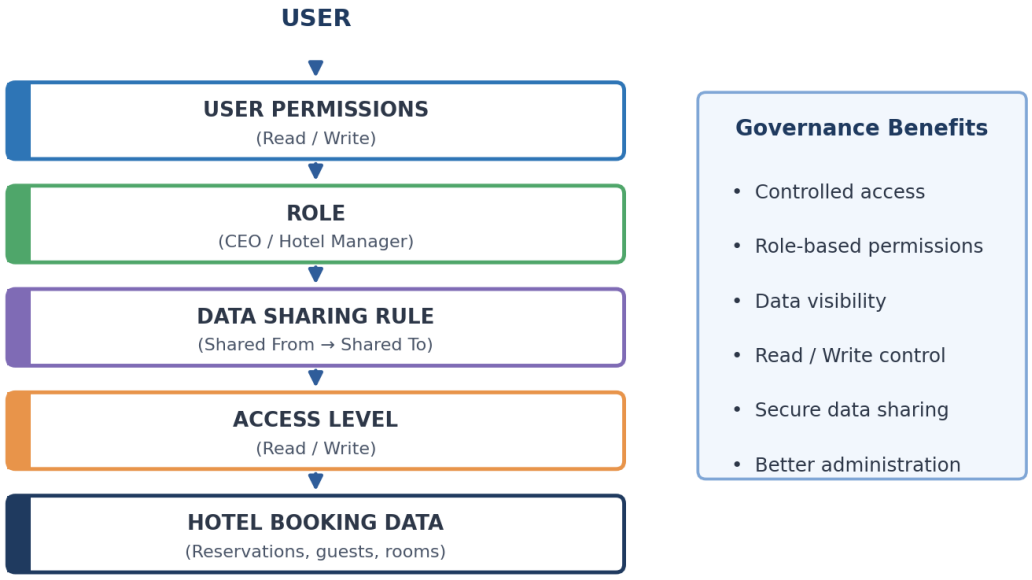


Figure 7: Role-Based Access and Data Sharing Flow (Conceptual Diagram)

- A user is first given permissions (Read / Write).
- The user belongs to a role, such as CEO or Hotel Manager, which fixes the position in the hierarchy.
- A data sharing rule specifies the records shared from one role or user to another.
- The access level (Read / Write) defines what the recipient can do with the Hotel Booking data.

Governance Best Practices

- **Least privilege:** give each user only the access needed for their work.
- **Use roles, not individuals:** assign access through roles and permission sets so it is easy to manage.
- **Share selectively:** create data sharing rules only where records must be visible beyond the hierarchy.
- **Separate external access:** manage guest access through portal permissions, not internal roles.
- **Review regularly:** check roles, permissions and sharing rules whenever staff or responsibilities change.

Governance Observations

- Predefined Read and Write permissions are available in the application.
- The Roles tab shows the role hierarchy with the default CEO role, and a role can be defined through Add Role.
- A data sharing rule can extend record visibility with a chosen access level.
- Portal User Permissions control external user access.
- The availability of some advanced security options depends on the subscription plan of the account.

CONCLUSION

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment, the predefined Read and Write permissions were examined, the role hierarchy was reviewed, a role was defined for the Hotel Booking application and a data sharing rule was configured to share booking records with a chosen access level. Some advanced governance features depend on the subscription plan. Controlling who can access application data, and at what level, remains an essential part of building and managing business applications.

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